

NRES 746 – Workout of the Day #1: Math & Stats Self-Diagnostic

This is **not graded and will not be collected**. There are no wrong answers – the goal is just to take stock of the tools you’re bringing into this class, and to see which ones might need a bit of dusting off. Some of these are meant to make you think a little! Keep this sheet; we’ll refer back to it later in the semester.

Instructions: Work alone for about 5 minutes. Don’t worry if you get stuck – just note where you got stuck and move on. After 5 minutes, compare answers with a neighbor for a few minutes.

Part 1: Logarithms & exponents _____

1a. Solve for x : $2^x = 32$

$x =$ _____

1b. A population grows according to $N_t = N_0 e^{rt}$. If $N_0 = 100$ and $N_t = 550$ after $t = 8$ years, solve for the growth rate r :

$r =$ _____

Part 2: Derivatives & integrals _____

2a. If $f(x) = (4x^2 - 3x)^5$, what is $f'(x)$?

$f'(x) =$ _____

2b. A recruitment rate for a population is given by $r(t) = 4t$ (new individuals per unit time). Compute the total number of new recruits between $t = 0$ and $t = 5$:

$\int_0^5 4t \, dt =$ _____

Part 3: Probability _____

3a. $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \text{ and } B) = 0.2$. What is $P(A \text{ or } B)$?

3b. A family has two children. You are told that at least one of them is a girl. What is the probability that *both* children are girls? Briefly show your reasoning (consider all the equally likely possibilities for two children).

3c. What if I reworded the above question... You are now told: A family has two children. You go to their home and observed a girl playing in the backyard. What is the probability that *both* children are girls? Does this wording change your answer from above?

Part 4: Confidence intervals & p-values ---

4a. Circle the interpretation you think is *most correct* for a 95% confidence interval on a mean:

- a) There is a 95% probability that the true mean falls within this particular interval.
- b) If we repeated this study many times, about 95% of the intervals we compute would contain the true mean.
- c) 95% of the data fall within this interval.

4b. Circle the definition you think is *most correct* for a p-value:

- a) The probability that the null hypothesis is true.
- b) The probability of observing results at least as ‘extreme’ under the null hypothesis.
- c) The probability that the alternative hypothesis is false.

Part 5: R / pseudocode ---

5. Write a function `GetCV()` that computes the coefficient of variation (standard deviation divided by the mean) of a numeric vector `x`. You can use `sd()` and `mean()`.

```
GetCV <- function(x){  
  
}
```

Self-rating ---

For each part above, circle how confident you felt (1 = totally lost, 5 = very confident):

Part	1	2	3	4	5
Logs & exponents	☐	☐	☐	☐	☐
Derivatives/integrals	☐	☐	☐	☐	☐
Probability	☐	☐	☐	☐	☐
Confidence intervals/p-values	☐	☐	☐	☐	☐
R / pseudocode	☐	☐	☐	☐	☐